

Pure Substances and Elements

A pure substance is made of one kind of particle; elements are pure substances that can't be broken down further.

Pure substances and elements

In science, a **pure substance** is made of only **one type of particle** and cannot be separated into other substances by any physical process. (This is stricter than the everyday meaning of "pure" = unadulterated.)

Elements

An **element** is a pure substance that **cannot be broken down into simpler substances**. Elements are the building blocks of all matter. Each element is made of identical particles called **atoms**, different from every other element's atoms.

- Examples: hydrogen, oxygen, carbon, sulfur, gold, silver, iron.
- Atoms often join to form a **molecule** (e.g. two hydrogen atoms → one hydrogen molecule).

Metals, non-metals and metalloids

Metals	Non-metals	Metalloids
Gold, silver, iron, magnesium, aluminium	Carbon, sulfur, hydrogen, oxygen	Silicon, boron (in-between)

Adulteration is illegally adding cheaper/poor substances to a product — a "pure" packet of food may still be a mixture to a scientist if it has more than one substance.

Where students lose marks

Trap 1 — Element = cannot be broken down further. Hydrogen and oxygen are elements; water is not (it splits into them).

Trap 2 — Atom vs molecule. An atom is the smallest particle of an element; a molecule is two or more atoms joined together.

Trap 3 — "Pure" in science ≠ "pure" on a label. Science: one kind of particle. Everyday: unadulterated.

Metalloids (silicon, boron) have properties between metals and non-metals.

Model answers — write them like this

Example A. Why is oxygen called an element but water is not?

Step 1: Oxygen cannot be broken into simpler substances.

Step 2: Water can be split (by electricity) into hydrogen and oxygen.

∴ Oxygen is an element; water is made of two elements.

Example B. Classify: gold, sulfur, silicon.

Step 1: Gold → metal; sulfur → non-metal.

Step 2: Silicon has in-between properties → metalloid.

∴ Gold = metal, sulfur = non-metal, silicon = metalloid.

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